HONEYWELL CAMPUS CONNECT

- **Problem Statement ID –** 1
- **Problem Statement Title -** AI-Powered Autonomous Smart Building Optimization Challenge
- **Theme –** Transforming Buildings with Autonomous AI and Digital Twins
- **PS Category -** Software
- **Student Name -** Utkarsh Dubey
- **Student ID -** 20484617

EcoLoop Building Agents

Autonomous AI-Powered Smart Building Optimization Engine

Proposed Solution

AI-Powered Building Management System

- 🏢 Energy Plus Digital Twin
- 🤖 Open-Source LLM (Llama 3.1 / Qwen)
- 🔗 Model Context Protocol (MCP)
- ⚡ Autonomous HVAC Optimization

Innovation & Uniqueness

- 🔄 Closed-Loop Autonomous Control
- 🔗 Energy Plus + MCP + LLM Integration
- 🧠 Explainable AI Decisions
- 🛡 Fault-Tolerant AI Recovery

How It Solves the Problem

- ✓ Predictive AI Control
- ✓ Optimizes Energy, Comfort & Carbon
- ✓ 2-Hour Occupancy Forecasting
- ✓ Confidence-Based Safety Layer

Continuously Monitors

- 🌡 Temperature
- 👥 Occupancy
- ⚡ Energy Demand
- 🌎 Grid Carbon Intensity
- ⏱ Every 15 Minutes

Performance Results

- 🟩 9.9% HVAC Energy Savings
- 🟩 14.2% Carbon Reduction
- 🟩 100% Thermal Comfort Compliance
- 🟩 Zero-Crash Fault Recovery

Validated On

- 🏢 DOE Reference Building
- 📏 ASHRAE 90.1 / 62.1
- 🌡 ISO 7730 PMV
- ⚡ EIA-930 Grid Carbon Data

Assess → Forecast → Trade-off → Decide

TECHNICAL APPROACH

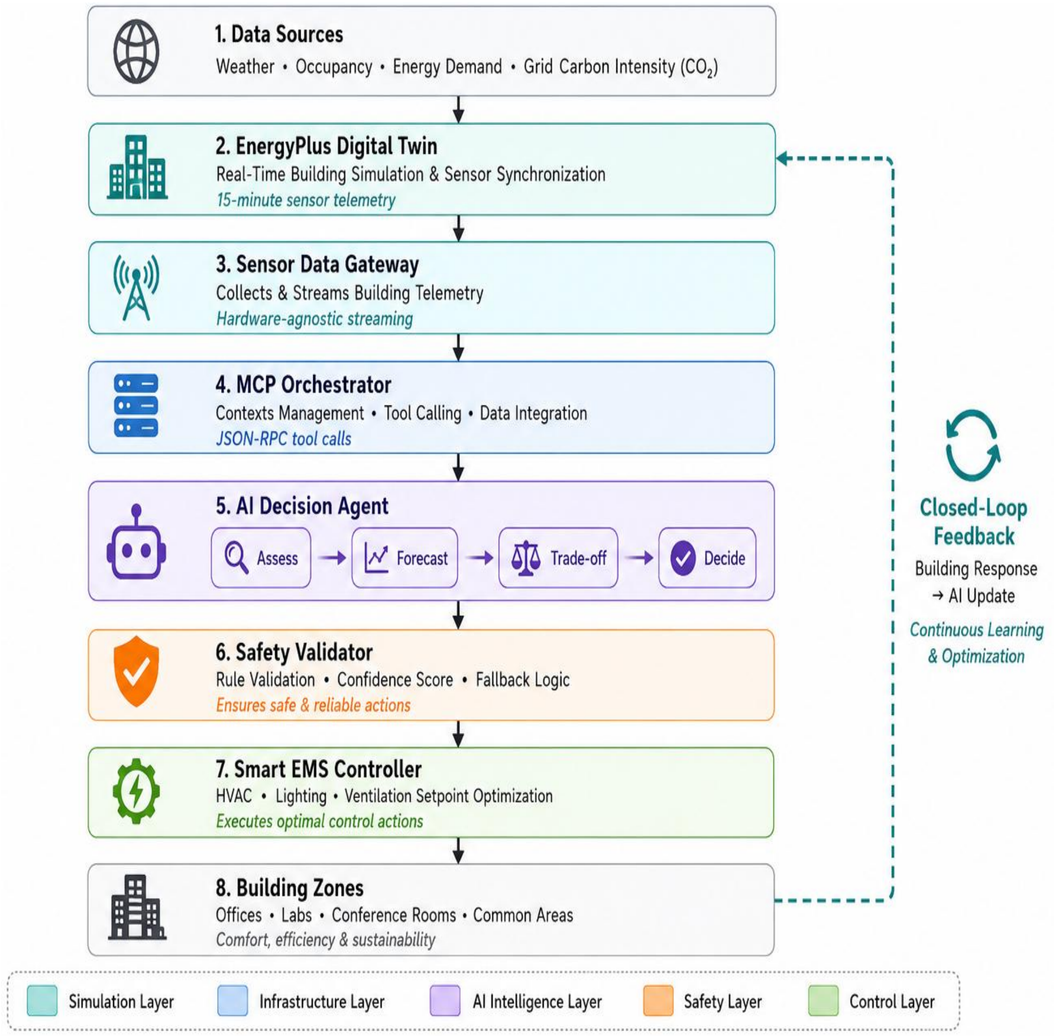


Figure 1. End-to-end closed-loop autonomous building control architecture showing data acquisition, AI reasoning, safety validation, and EnergyPlus-based HVAC control

Physics-Based Digital Twin: Simulates realistic building behavior using EnergyPlus.

Real-Time Data Integration: Combines weather, carbon, and occupancy telemetry.

AI-Driven Optimization: Predicts and optimizes HVAC setpoints using MCP-enabled reasoning.

Safety-First Control: Validates every AI action against comfort and operational constrain

Closed-Loop Feedback: Continuously monitors outcomes to refine future decisions

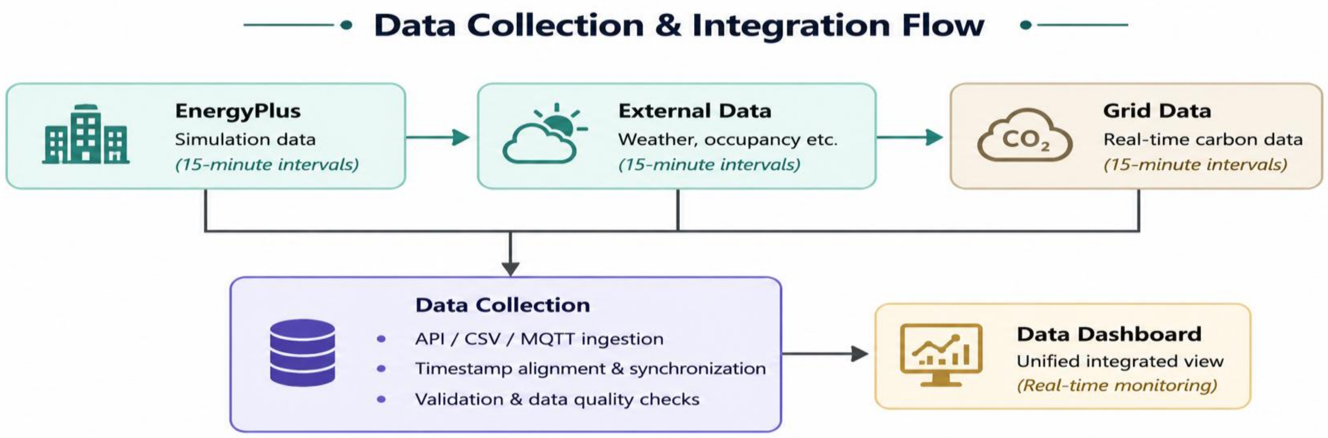


Figure 2. Data lineage pipeline illustrating how raw EnergyPlus simulation telemetry is processed into validated engineering metrics and visualized on the dashboard.

FEASIBILITY AND VIABILITY

Feasibility Analysis

. Built entirely using **open-source technologies** (EnergyPlus, Python, Ollama, Streamlit) with **no cloud or proprietary dependencies**.

. Validated on the **DOE Commercial Reference Building** (ASHRAE 90.1 compliant) using real EnergyPlus simulation.

. Uses **real EIA-930 historical grid carbon data** and ISO 7730 PMV comfort model for realistic evaluation.

Potential Challenges & Risks

. LLM inference latency may affect large-scale real-time deployments.

. Noisy or faulty sensor data can lead to incorrect control decisions.

. Legacy Building Management Systems may lack modern BACnet/IoT interfaces.

Scaling from simulation to large multi-building environments requires additional deployment infrastructure.

Strategies to Overcome Challenges

🛡️ Confidence-Based Safety Layer

🔍 AI-Based Anomaly Detection

🔗 Hardware-Agnostic Gateway

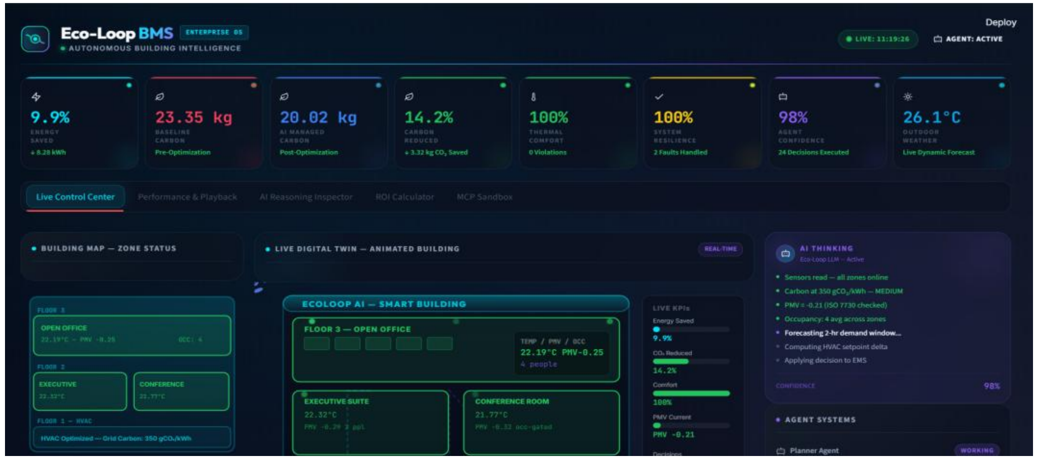
🚀 Incremental Deployment

🧪 Automated & Stress Testing

📊 Continuous Performance Monitoring

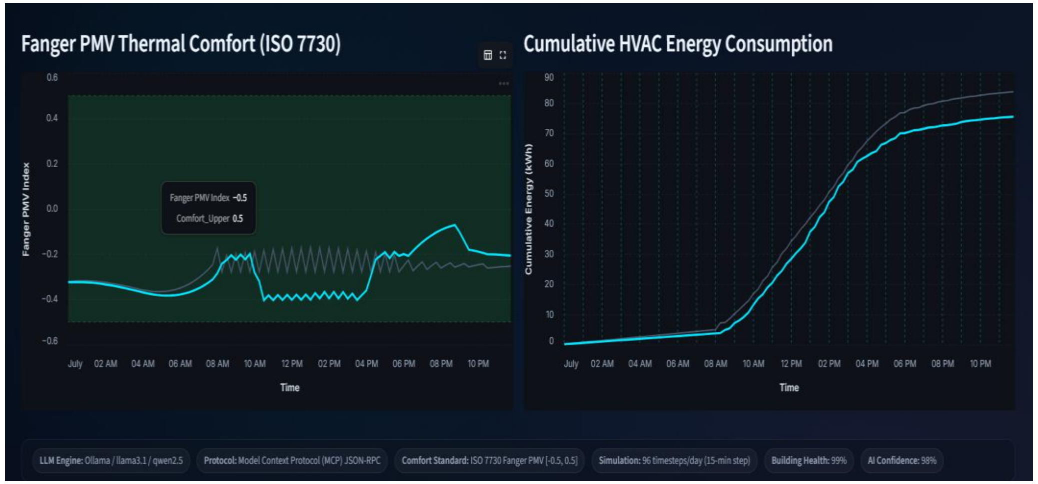
🔄 Rule-Based Fallback Control

ARTIFACTS

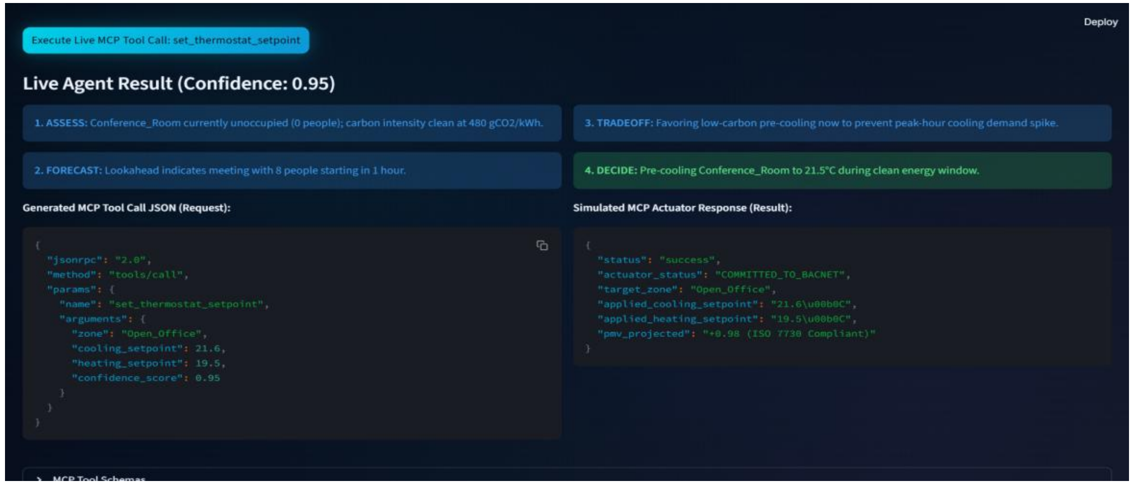


Live AI Control Dashboard

Explainable AI Decision Engine



Performance Benchmark & Validation



Model Context Protocol (MCP) Integration

RESEARCH AND REFERENCES

Standards & Building Models

Simulation Platform

EnergyPlus v26 (DOE) – Building Energy Simulation

DOE Commercial Reference Building – Small Office Benchmark

Building Standards

ASHRAE 90.1 – Energy Efficiency

ASHRAE 62.1 – Indoor Ventilation

ISO 7730 (PMV) – Thermal Comfort Model

Datasets & Benchmark Sources

Weather & Energy

Chicago TMY3 EPW – Historical Weather Data

EIA-930 – Hourly Grid Demand & Carbon Data

PJM ComEd Carbon Intensity – Historical Carbon Emissions

Benchmark Dataset

DOE CBECS Database – Commercial Building Energy Benchmark